

Bounding Decision Equivalence in Encrypted Face Verification under RNS-CKKS

Qiyue Zhao

Abstract. A face template cannot be revoked, so a verification service should match against templates that it is unable to read. Homomorphic encryption meets this requirement: the client encrypts its template, and the server scores it while holding no decryption key. What decides whether such a service is deployable is whether encrypted arithmetic ever changes an accept-or-reject outcome. Existing encrypted matchers answer this question by counting the outcomes that changed in one execution, and a count of zero describes the noise that the execution happened to draw rather than a property of the system. In this paper, we replace that count with an inequality. A decision depends on a score only through the sign of its distance to the threshold, so the two pipelines agree whenever the largest score error is smaller than the smallest such distance. We make the latter quantity positive by construction, by placing every candidate threshold at a midpoint of the observed score grid, and we report the ratio between the two. The same view fixes the precision of the scoring circuit without a fitted constant and yields a lower bound on the scaling factor that a deployment may use. On the 6,000 pairs of the Labeled Faces in the Wild pair protocol, an RNS-CKKS pipeline at multiplicative depth one reaches 99.87% accuracy, with a largest score error of 1.26×10^{-6} against a smallest margin of 2.08×10^{-3} , a ratio of 6.07×10^{-4} . One verification costs 5.75 ms of server time, which falls to 0.73 ms when eight templates share a ciphertext.

Keywords: Biometric template protection · Homomorphic encryption · RNS-CKKS · Face verification · Approximate arithmetic.

1 Introduction

A password can be replaced after a breach, whereas a face cannot. A template extracted by a modern recognition network supports the reconstruction of a recognisable image and permits linkage across databases [14,17], so a service that stores templates in the clear holds a permanent identifier for every person it has enrolled. This is the reason that matching should take place on data that the matching party is unable to read.

Homomorphic encryption provides exactly this separation. The client encrypts its template, the server evaluates the comparison on ciphertexts, and the server holds evaluation keys alone. For embeddings of unit Euclidean norm the comparison is a cosine similarity, which is an inner product, so a CKKS matcher [5] evaluates it with one ciphertext multiplication and a logarithmic

number of rotations. Encrypted face matching along these lines is established for verification against a single template [3] and has been extended to search over a gallery [8,9].

CKKS is nevertheless an approximate scheme, so the score that the client decrypts differs from the score that the cleartext pipeline would have produced. Whether this difference matters is the question a deployment actually asks, because a verification service delivers decisions rather than scores. Existing systems answer it empirically: they execute both pipelines, compare the two sets of decisions and report that none of them changed. Such a count, however, describes a single execution. The noise that a ciphertext carries depends on the key set that was drawn, and a pair whose score lies arbitrarily close to its threshold will change its outcome under an arbitrarily small perturbation. A count of zero therefore leaves open the two questions that matter for deployment: how close the system came to a change, and whether a different key set would produce one.

Our starting point is that a decision depends on the score only through the sign of the difference between the score and the threshold. Consequently, the encrypted and the cleartext pipelines agree on a given pair whenever the error of that pair’s score is smaller than the distance from its cleartext score to the threshold. Writing E for the largest error over the evaluated pairs and M for the smallest such distance, the single inequality $E < M$ is sufficient for agreement on every pair at once, whatever noise the execution drew. This inequality is informative only if M is strictly positive, and M is strictly positive only if no score coincides with a threshold. We obtain that by construction rather than by chance. The candidate thresholds of each fold are the midpoints between adjacent distinct training scores, so no candidate can equal an observed value. Reporting the ratio E/M then replaces a count of changed decisions with a margin, and states how much room the arithmetic retains before any decision is at risk.

The same view turns the precision of the scoring circuit into a quantity worth predicting rather than merely measuring. A CKKS plaintext carries the message multiplied by the scaling factor, and the ciphertext noise is governed by the widths of the moduli and by the number of key switches instead of by that factor. Each additional scale bit therefore buys exactly one precision bit, which fixes the slope of precision against scale bits at one and leaves no constant free to be tuned against the data. Combining this model with the measured margin yields a design rule: a deployment may reduce the scaling factor until the resulting error reaches M , and the number of bits between the working point and that limit is the precision that may be traded for a shorter modulus chain.

To summarise, this paper makes the following contributions.

- This project formulates the agreement between an encrypted matcher and its cleartext reference as a margin inequality instead of a count of changed decisions, and we make the margin strictly positive by construction through a midpoint threshold grid. On the Labeled Faces in the Wild pair protocol the two pipelines agree on all 6,000 pairs, and the ratio of the largest score

error to the smallest margin is 6.07×10^{-4} , so the agreement holds with a factor of 1,647 to spare and does not rest on the noise that this execution drew.

- We give a parameter-free account of the scoring circuit. The precision model predicts a slope of exactly one against the scale bits, and the sweep measures 0.957 with a coefficient of determination of 0.9880. The rotation schedule of the fold predicts a speed ratio of 56.8 and a noise ratio of 256 against a naive single-slot fold, and the measurements give 57.2 and 332. Neither prediction contains a constant fitted to the data.
- The project instantiates the design at multiplicative depth one with Microsoft SEAL and evaluate it on real photographs. The encrypted pipeline reaches 99.87% accuracy under ten-fold cross-validation, one verification costs 5.75 ms of server time and 257.1 KiB of uplink, and packing eight templates into one ciphertext reduces these to 0.73 ms and 32.1 KiB. Each operating point declares whether the impostor sample can express the false accept rate it quotes, and no value is reported where the sample cannot.

2 Related Work

Encrypted Face Matching. Boddeti [3] established matching between templates encrypted under a fully homomorphic scheme and reported the template size and the per-match cost that follow from packing a 512-dimensional feature into one ciphertext. Drozdowski et al. [8] applied the same primitive to identification, and Engelsma et al. [9] addressed the regime in which the cost of scanning a gallery dominates and the encoding of that gallery becomes the design problem. All three report accuracy figures that match their cleartext references, and all three establish the match by comparing measured rates. The present work keeps to the one-to-one case and asks what supports such a claim.

The Approximation Error of CKKS. CKKS [5] and its full residue number system variant [4] are approximate by design, and the widths that a security level permits are tabulated by the Homomorphic Encryption Standard [2]. The precision of a CKKS computation has accordingly been analysed at the level of the scheme, in terms of the noise that encoding, multiplication and key switching contribute. What such analyses do not supply is the step from the precision of a circuit to the correctness of the decision that a task takes on its output, which requires a quantity drawn from the task rather than from the scheme.

Deciding under Encryption. A matcher may also compare against the threshold homomorphically and return only the resulting bit. Doing so requires a polynomial approximation to the sign function [6,13], whose multiplicative depth is substantially larger than that of the inner product and whose cost is highest for the scores that lie closest to the threshold. We deliberately exclude that depth and state the consequence in the trust model of Section 3.1: the

server learns nothing about the ciphertexts it processes, while the client learns the numeric score.

Protecting the Stored Template. Cancelable transforms and biometric cryptosystems enrol a transformed template so that a compromised database does not disclose the original [16,17]. They protect storage rather than the computation performed during a match, and encryption is complementary to them: it removes the server from the trust boundary while the match is taking place.

What the Released Score Discloses. A released score is itself a disclosure channel. Hill-climbing attacks recover a template from repeated score queries [1,10], and face images can be reconstructed from deep templates [14]. Encryption does not prevent either, because both exploit the decrypted output. Bounding that leakage is a separate problem from the one studied here, and Section 5 returns to it.

None of these lines of work states, for the pairs on which a system actually decides, how far the encrypted arithmetic stands from changing a decision. That is the quantity this paper supplies.

3 The Scoring Circuit and Its Margin

This section presents the system and trust model (Section 3.1), the templates and the circuit that scores them (Section 3.2), the margin bound on decision equivalence (Section 3.3), the precision model of the circuit (Section 3.4) and the amortisation that slot packing provides (Section 3.5).

3.1 System and Trust Model

System Model. A client holds a probe image and the secret key, and a server holds the enrolled templates together with the relinearization and Galois keys alone. Enrolment sends one encrypted template, which the server stores. Verification sends one encrypted probe; the server returns one encrypted score, and the client decrypts that score and compares it with the threshold of its fold. Fig. 1 places the four stages against the trust boundary.

Trust Model. The server is honest but curious. It observes ciphertexts and the sizes and timings of the operations it performs, and it holds no key with which to decrypt them. The client holds the secret key and therefore learns the numeric score of every verification it requests. The separation is enforced at build time: the server library and the server executable are compiled without the translation unit that constructs a secret key, so no code path on the server side can reach a decryption routine.

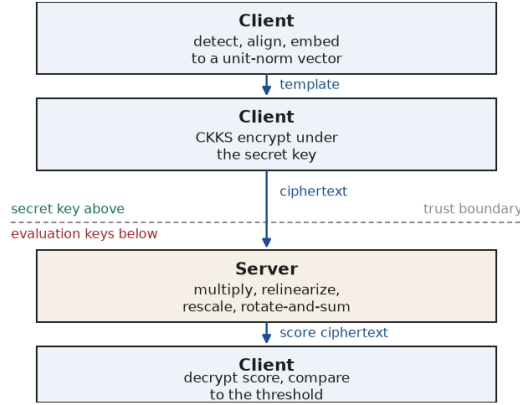


Fig. 1. The verification pipeline. The secret key remains on the client, and the server holds evaluation keys alone.

Why the Client Decides. Comparing a score against a threshold under encryption requires a polynomial approximation to a step function, and the multiplicative depth that such an approximation consumes is an order of magnitude greater than the depth of the comparison itself [6,13]. Excluding it keeps the modulus chain at three primes and the circuit at depth one, which is what makes the costs of Section 4.4 as small as they are. The price is stated rather than hidden: a deployment in which the client should learn only the accept-or-reject bit requires either a deeper circuit or a second non-colluding party, and neither is treated here.

3.2 Templates and the Scoring Circuit

Each photograph passes through three stages before encryption. A detector predicts a bounding box together with five landmarks, namely the two eye centres, the nose tip and the two mouth corners. A similarity transform, estimated in the least-squares sense over those five points, maps them onto the canonical five-point template of ArcFace and produces a 112×112 crop in which the eyes and the mouth occupy fixed pixel positions. The recognition network maps that crop to a vector of dimension $m = 512$, which is then scaled to unit Euclidean norm.

Normalisation Stays on the Client. A square root and a division are expensive under homomorphic encryption, whereas both are free in the clear. Performing them on the client leaves the server with an inner product and is what keeps the circuit at depth one. The cost of this choice is that the server cannot verify that a probe it receives has unit norm, which Section 5 discusses. Its benefit is that, for templates x and g of unit norm, the quantity the server computes is

Table 1. The RNS-CKKS parameter set. The widths follow from the circuit: one ciphertext product squares the scale, and the single rescale divides it by the middle prime and restores it.

Context	Value	Circuit	Value
Polynomial modulus degree N	8,192	Multiplicative depth	1
Plaintext slots	4,096	Rotations per score	9
Coefficient modulus (bits)	60, 40, 60	Slots per template	512
Total modulus width	160 of 218	Templates per ciphertext	8
Scaling factor Δ	2^{40}	Security level	128-bit classical

exactly the cosine similarity

$$s = \langle x, g \rangle \in [-1, 1]. \quad (1)$$

The parameters of the scheme follow from this circuit and are listed in Table 1. A polynomial modulus degree of $N = 8192$ gives $N/2 = 4096$ plaintext slots. The coefficient modulus is a chain of primes of 60, 40 and 60 bits, which totals 160 bits against the 218 that the 128-bit classical security level permits at this degree [2]. The last prime in the chain is the key-switching prime and carries no plaintext, so the data chain holds two levels and the parameter set admits exactly one multiplication. The scaling factor is $\Delta = 2^{40}$.

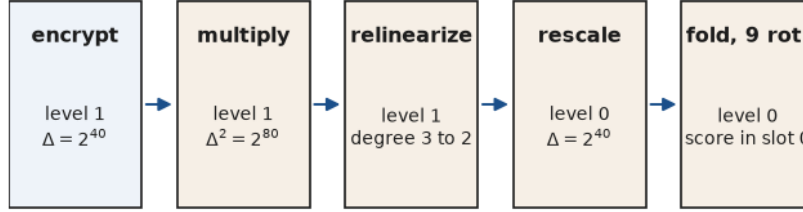
The server holds two ciphertexts, each carrying a template in the low slots of its block, and evaluates four operations. It multiplies the two ciphertexts, which yields the slot-wise products and a ciphertext of degree three. It relinearizes, returning the ciphertext to degree two. It rescales, which restores the scale and consumes the one available level. It then folds the products into a single slot with nine rotations. Fig. 2 records the chain index and the scale after each step.

The Fold Uses Doubling Strides. Adding to a ciphertext a copy of itself rotated by one slot leaves every slot holding the sum of two neighbours. Repeating with strides of two, four and so on up to $d/2$ leaves slot i holding the sum of the d consecutive slots that begin at i , where $d = 512$ is the block length. Since d is a power of two and divides the slot count, the window that begins at a block boundary covers exactly that block, and the first slot of each block therefore holds the inner product of the template it carries. Galois keys are generated for those nine strides alone, which is why the Galois key measures 7,005.3 KiB rather than the far larger key that arbitrary rotations would require. Proposition 4 quantifies what this schedule buys against the alternative of rotating by a single slot repeatedly.

3.3 Decision Equivalence as a Margin

A verification service delivers decisions, so the property that matters is not the proximity of the two score sets but the identity of the two decision sets. We state that property and give a sufficient condition for it.

multiplicative depth one: the single rescale below is the only level spent



modulus chain 60, 40 bits, key-switching prime 60 bits

Fig. 2. The scoring circuit, with the chain index and the scale after each step. The single rescale is the only level that the parameter set is able to spend.

Definition 1 (Decision equivalence). Let P be a set of pairs in which each pair p carries a cleartext score s_p , an encrypted score $\hat{s}_p$ and a threshold τ_p . The cleartext pipeline decides $D(p) = \mathbf{1}[s_p \geq \tau_p]$ and the encrypted pipeline decides $\hat{D}(p) = \mathbf{1}[\hat{s}_p \geq \tau_p]$. The two pipelines are decision-equivalent on P if $D(p) = \hat{D}(p)$ for every $p \in P$.

Proposition 1 (Margin bound). Let $E = \max_{p \in P} |\hat{s}_p - s_p|$ and $M = \min_{p \in P} |s_p - \tau_p|$. If $E < M$, the two pipelines are decision-equivalent on P .

Proof. Fix $p \in P$ and write $e = \hat{s}_p - s_p$, so that $|e| \leq E$. Suppose first that $s_p \geq \tau_p$. Then $s_p - \tau_p = |s_p - \tau_p| \geq M > E \geq -e$, whence $\hat{s}_p - \tau_p = (s_p - \tau_p) + e > 0$ and $\hat{D}(p) = 1 = D(p)$. Suppose next that $s_p < \tau_p$. Then $\tau_p - s_p \geq M > E \geq e$, whence $\hat{s}_p - \tau_p = e - (\tau_p - s_p) < 0$ and $\hat{D}(p) = 0 = D(p)$. Since p was arbitrary, the two decisions agree on every pair of P .

The condition is an inequality between two scalars, one supplied by the arithmetic and one supplied by the task. It is also uniform over P : a single comparison settles every pair at once, and it does so for any perturbation bounded by E rather than for the particular perturbation that one execution produced. Its content, however, depends entirely on M being strictly positive, and M is a property of how the thresholds were chosen.

Proposition 2 (A midpoint grid keeps the margin positive). Let $u_1 < u_2 < \dots < u_n$ be the distinct cleartext scores of the split on which the threshold of a fold is fitted, let $\delta > 0$, and let the candidate set be

$$\Theta = \{u_1 - \delta\} \cup \left\{ \frac{1}{2}(u_{i-1} + u_i) : 2 \leq i \leq n \right\} \cup \{u_n + \delta\}. \quad (2)$$

Then no element of Θ equals a score of that split, so every pair of the split retains a strictly positive distance from any threshold selected from Θ .

Proof. A midpoint satisfies $u_{i-1} < \frac{1}{2}(u_{i-1} + u_i) < u_i$ because $u_{i-1} < u_i$, and the values $u_1, \dots, u_n$ are the only scores of the split, so the midpoint coincides with none of them. The two sentinels lie outside $[u_1, u_n]$ and therefore coincide with none of them either.

Why a Midpoint Grid Rather than the Scores Themselves. The obvious alternative is to search over the observed scores, which is how a threshold grid is often built. Such a grid places at least one pair at distance zero from the threshold by construction. The ratio E/M is then unbounded, Proposition 1 has nothing to say, and a count of zero changed decisions rests on the direction in which the noise of that pair happened to point. The midpoint grid removes this systematic case at no cost in accuracy, because a threshold placed between two adjacent distinct scores classifies the split exactly as one placed at either of them. It does not remove the accidental case in which a held-out score coincides with the threshold that is applied to it, which is why M is measured over all 6,000 pairs in Section 4.3 rather than asserted.

3.4 Precision of the Scoring Circuit

Proposition 1 makes E the quantity that the arithmetic must keep small, so it is worth knowing what governs it. The answer is a one-line model with no room in it for a fitted constant, which makes it falsifiable against the sweep of Section 4.6.

Proposition 3 (One scale bit buys one precision bit). *Consider the circuit of Fig. 2 evaluated at scaling factor Δ , with the widths of the primes in the modulus chain held fixed, and let ν denote the ciphertext noise that the circuit accumulates in the coefficient domain. If ν is independent of Δ , the absolute error of the decoded score is $\varepsilon = \nu/\Delta$ and the precision in bits satisfies*

$$-\log_2 |\varepsilon| = \log_2 \Delta - \log_2 \nu, \quad (3)$$

so precision is an affine function of the scale bits with slope exactly one and with no free constant, the intercept $-\log_2 \nu$ being the noise floor of the circuit.

Proof. A CKKS plaintext encodes a message m as $\lfloor \Delta m \rfloor$, and decoding divides by Δ . The contributions that encoding, key switching and rounding make to the ciphertext enter additively in the coefficient domain and are governed by the widths of the moduli and by the number of key switches, neither of which involves Δ . The rescale that follows the multiplication divides the squared scale and the accumulated noise by the same middle prime, so it restores Δ and leaves the ratio ν/Δ unchanged. Decoding therefore returns $m + \nu/\Delta$, and taking the negative base-two logarithm of the second term gives (3).

The Assumption Is First Order. The noise that a multiplication contributes is proportional to the magnitude of its operands and hence to Δ , and it is the rescale that removes this proportionality; the rounding introduced by encoding is bounded independently of Δ . Treating ν as constant is therefore an approximation, and the deviation of the measured slope from one is the residual of that approximation rather than a fitted quantity. Section 4.6 measures 0.957, a deviation of -0.043 .

Proposition 4 (Cost of the two folds). *Let d be a power of two and let a block of d slots hold one template. The doubling fold, which adds copies rotated by $1, 2, \dots, d/2$, produces the block sum with $\log_2 d$ rotations; the single-slot fold, which adds a copy rotated by one slot repeatedly, needs $d - 1$. Counting one unit of noise per key switch and attributing to each term the number of rotations it has undergone, the single-slot fold accumulates $d(d - 1)/2$ units against $d - 1$ for the doubling fold. The two ratios are therefore*

$$\rho_{\text{time}} = \frac{d - 1}{\log_2 d}, \quad \rho_{\text{noise}} = \frac{d}{2}. \quad (4)$$

Proof. The rotation counts are immediate: the doubling fold has $\log_2 d$ steps by construction, and the single-slot fold must bring each of the remaining $d - 1$ slots into position. For the noise, observe that in the single-slot chain the k -th term reaches the accumulator after k rotations and therefore carries k units, so the chain accumulates $\sum_{k=1}^{d-1} k = d(d - 1)/2$. In the doubling fold, if the ciphertext entering step j carries a_j units then the rotated copy carries $a_j + 1$, so $a_{j+1} = 2a_j + 1$ with $a_0 = 0$; hence $a_j = 2^j - 1$ and the fold ends at $a_{\log_2 d} = d - 1$. Dividing the two pairs of counts gives (4).

For $d = 512$ the predicted ratios are 56.8 and 256. Section 4.4 measures 57.2 and 332 for the same two folds on the same host, so the schedule is favourable on both axes and by the predicted amounts. The two effects carry different exponents, which is what makes the choice of schedule worth stating: the speed ratio is linear against logarithmic, whereas the noise ratio is quadratic against linear.

3.5 Slot-Packed Batching

A block of $d = 512$ slots holds one template and a ciphertext at $N = 8192$ carries 4,096 slots, so $B = 8$ templates occupy one ciphertext side by side. The multiplication is slot-wise and, by the argument of Section 3.2, the fold is block-local, so a single pass of the circuit produces B independent scores, each in the first slot of its own block. Fig. 3 shows the occupancy before and after the fold.

Since the circuit is the same at every batch size, its latency and its uplink are divided by B when they are charged to a single verification. A service under load therefore amortises one pass over eight comparisons, which is the regime that Section 4.5 measures. Packing does not alter the argument of Section 3.3, because the fold is block-local and each score is produced by the same sequence of operations as in the unbatched circuit.

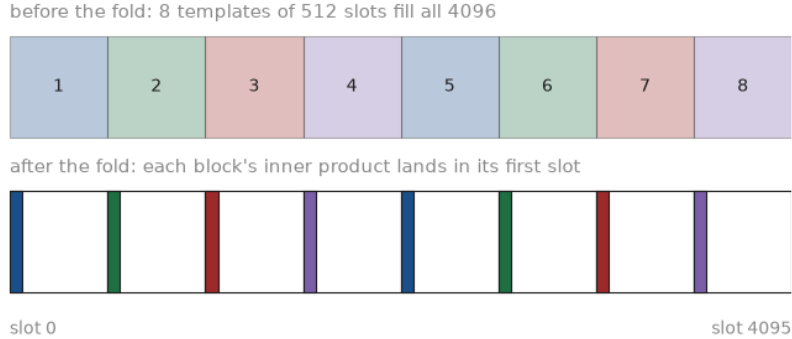


Fig. 3. Slot occupancy of one ciphertext before and after the fold. Each template holds a block, and the fold leaves that block’s inner product in the block’s first slot.

4 Experimental Evaluation

The evaluation answers four questions. Section 4.2 establishes whether the encrypted pipeline decides as well as its cleartext reference. Section 4.3 quantifies the room left by Proposition 1. Sections 4.4 and 4.5 report the cost of one verification and the gains obtained from the fold schedule of Proposition 4. Section 4.6 verifies that the precision model of Proposition 3 holds and derives the implications for the scaling factor a deployment should choose.

4.1 Dataset, Protocol and Setup

Dataset and Protocol. Labeled Faces in the Wild [12] fixes 6,000 pairs in 10 folds, each fold holding pairs of the same person and pairs of different people in equal number. This evaluation uses the View-2 pair protocol with 3,000 genuine pairs and 3,000 impostor pairs over 7,701 deep-funnelled photographs. The fold index of every pair is carried from the protocol file into the evaluation, so the accuracy reported below is comparable with published figures for the same split.

Template Frontend. Faces are located by the SCRFD detector [11] at an input resolution of 320 pixels, aligned onto the canonical five-point ArcFace template by a similarity transform, and embedded into 512 dimensions by a ResNet-50 trained with the ArcFace loss [7] on WebFace-600K [18]. The detector located a face in all 7,701 images, and the largest deviation of any template from unit norm is 2.220×10^{-16} , so the inner product of (1) is the cosine similarity to within the precision of the double-precision reference itself.

Thresholds. For each fold the threshold maximises accuracy on the other nine folds and is then applied to the held-out fold. The candidate set is the grid of Proposition 2, built from the scores of the nine training folds.

Table 2. Verification accuracy under the fold protocol. Each threshold is fitted without the fold to which it is applied.

Quantity	Cleartext	Encrypted
Accuracy, mean over folds	99.87%	99.87%
Accuracy, standard deviation	0.19%	0.19%
Area under the ROC curve	0.999417	0.999417
Equal error rate	0.267%	0.267%
Threshold, mean over folds	0.2442	0.2442
Separation of the distributions	7.99	7.99

Cryptographic Backend and Host. The circuit is implemented with Microsoft SEAL 4.3.3 [15] under the parameters of Table 1, and serialization uses zstd compression. All measurements were taken on an Apple M1 with eight cores and 8 GiB of memory under a single-threaded evaluator, so the latencies below describe that machine and scale with its clock and memory system.

4.2 Verification Accuracy

Table 2 reports the ten-fold accuracy of both pipelines. The encrypted pipeline reaches 99.87% with a standard deviation of 0.19% across folds, against 99.87% with the same standard deviation for the cleartext pipeline, and the difference between the two is 0.0000%. The two pipelines also agree on the area under the receiver operating characteristic, 0.999417, on the equal error rate, 0.267%, and on the mean fitted threshold, 0.2442. The separation between the genuine and impostor score distributions, measured as the difference of their means over their pooled standard deviation, is 7.99. Fig. 4 shows the two distributions together with the ten fitted thresholds; the fold thresholds are visually indistinguishable from one another and sit in the empty interval that separates the two modes, which is what makes the margins of Section 4.3 as large as they are.

What the Sample Is Able to Express. A true accept rate quoted at a false accept rate of f requires at least one impostor pair per f of the sample, so 3,000 impostor pairs resolve no rate finer than 3.33×10^{-4} . Table 3 therefore carries two flags with every operating point: one recording whether the sample can express the requested rate at all, and one recording whether at least five false accepts support the estimate. At 10^{-2} the pipeline accepts 99.83% of genuine pairs and the estimate rests on 30 false accepts. At 10^{-3} it accepts 99.73%, but only three impostor pairs lie above the threshold, so the figure describes the position of those three pairs in the score ordering as much as it describes a rate. At 10^{-4} no value is reported, because a rate of that magnitude requires at least 10,000 impostor pairs and this sample holds 3,000. Reporting a number there would carry no information about the rate it names.

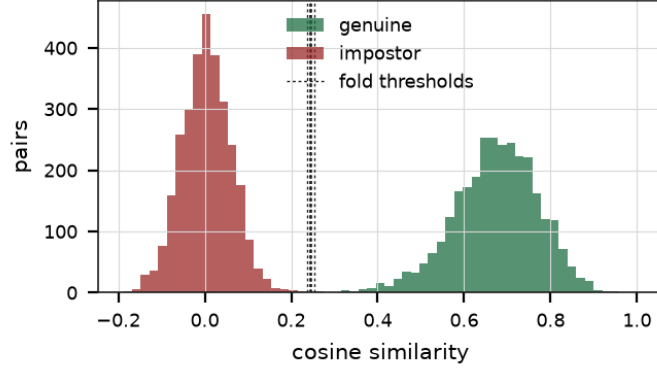


Fig. 4. Cleartext score distributions with the ten fitted fold thresholds.

Table 3. Operating points on the cleartext scores. A rate that the impostor count is unable to express carries no value.

Target rate	True accept	Achieved	False accepts	Supported
10^{-2}	99.83%	1.00×10^{-2}	30	yes
10^{-3}	99.73%	1.00×10^{-3}	3	coarse
10^{-4}	not reported	—	—	no

4.3 Numerical Fidelity and the Margin

Across all 6,000 pairs, the difference between an encrypted score and its cleartext reference has a mean absolute value of 2.01×10^{-7} , a median of 9.81×10^{-8} , a 99th percentile of 1.09×10^{-6} and a maximum of $E = 1.26 \times 10^{-6}$. The maximum corresponds to 19.6 bits of precision, which is within 0.4 bits of what the parameter set makes available: the surviving prime is 60 bits wide and carries a scale of 40 bits, which leaves 20 bits before the accumulated noise of the multiplication and the fold is subtracted. The circuit therefore spends almost none of its headroom, and the small shortfall is the noise floor that Section 4.6 measures directly.

The smallest distance from a cleartext score to the threshold of its fold is $M = 2.08 \times 10^{-3}$, so the ratio of Proposition 1 is

$$E/M = \frac{1.26 \times 10^{-6}}{2.08 \times 10^{-3}} = 6.07 \times 10^{-4}, \quad (5)$$

which lies below one by a factor of 1,647. The hypothesis of Proposition 1 is therefore satisfied with a wide margin, and the equality of the two decision sets follows from the inequality rather than from the particular noise this execution drew. The observed count of changed decisions is 0, which is what the bound requires and no longer what the claim rests on. Fig. 5 places the two quantities on

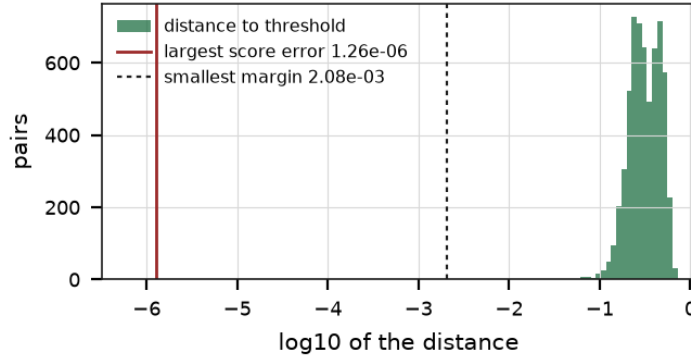


Fig. 5. Distances from each cleartext score to the threshold of its fold, against the largest score error. The gap between the two is the margin of Proposition 1.

one logarithmic axis, where the separation between the distribution of margins and the largest error is the bound itself.

Three Orders of Magnitude, and Where They Come From. The gap is large because two independent effects push in the same direction. The embedding separates genuine from impostor pairs well enough that the fitted thresholds fall in an interval that almost no pair occupies, which makes M of the order of 10^{-3} . The circuit spends one level and nine key switches, which places E of the order of 10^{-6} . Neither quantity is tuned against the other, and the ratio would have to grow by more than three orders of magnitude before a decision came under threat. Section 4.6 identifies the parameter that would be the first to bring it there.

4.4 Cost of One Verification

Table 4 reports medians over twenty unbatched repetitions. Server time for the circuit in which both templates are encrypted is 5.75 ms, of which the fold accounts for 4.41 ms, or 77% of the total; the end-to-end figure, which adds client encryption and decryption, is 10.52 ms. Fig. 6 gives the share that each stage takes. The asymmetry is worth noting: the multiplication that the parameter set was designed around costs 0.13 ms, and the summation that follows it costs thirty-four times as much. Uplink is 257.1 KiB for the probe ciphertext and downlink is 128.6 KiB for the score, while the relinearization and Galois keys, at 771.3 KiB and 7,005.3 KiB, are sent once per client and are not charged to a verification.

The Fold Schedule, Measured against Proposition 4. Replacing the doubling fold with one that rotates by a single slot 511 times raises server time from

Table 4. Cost of one unbatched verification with both templates encrypted. Times are medians; sizes are serialised bytes.

Stage	Cost	Note
Key generation	35.22 ms	once per client
Client encryption	4.57 ms	two templates
Server multiply	0.13 ms	depth 1
Server relinearize	0.93 ms	one key switch
Server rescale	0.27 ms	level 1 to 0
Server rotate-and-sum	4.41 ms	9 rotations
Server total	5.75 ms	
Client decryption	0.21 ms	one ciphertext
Uplink per verification	257.1 KiB	probe ciphertext
Downlink per verification	128.6 KiB	score ciphertext
Relinearization key	771.3 KiB	sent once
Galois key	7,005.3 KiB	9 strides, sent once

5.75 ms to 253.73 ms. Since the stages outside the fold are unchanged and account for 1.34 ms, the fold itself grows from 4.41 ms to 252.39 ms, a ratio of 57.2 against the 56.8 that ρ_{time} predicts for $d = 512$. Over the same repetitions the largest score error grows from 3.74×10^{-7} to 1.24×10^{-4} , a ratio of 332 against the 256 that ρ_{noise} predicts. The timing prediction is met to within 1%, and the noise prediction to within a factor of 1.3, the excess being the rounding that the longer chain accumulates and that a count of key switches does not capture. The consequence for the margin is direct: at 1.24×10^{-4} the ratio E/M would still be 0.060, so even the naive fold would satisfy Proposition 1, but it would do so with a factor of 17 in hand instead of 1,647.

The Price of Encrypting the Gallery. Keeping the enrolled template in the clear removes the relinearization from the critical path and brings server time to 4.74 ms, a saving of 18%. That saving is the price of the privacy property rather than a free optimisation: in the cleartext-gallery variant the server holds the enrolled template and therefore learns the identity it stores, which is the disclosure the system exists to prevent.

4.5 Slot-Packed Batching

Server time remains close to constant as the batch grows, because the circuit is the same at every batch size: 5.85 ms at a batch of one and 5.86 ms at a batch of eight. Amortised over the batch, this is 0.73 ms and 32.1 KiB of uplink per verification, a throughput of 1,364 verifications per second of server time. Table 5 gives every batch size. The largest score error rises from 1.54×10^{-6} at a batch of one to 1.70×10^{-6} at the full batch, an increase of 10% that lies inside the spread which independent executions show at this parameter set, so

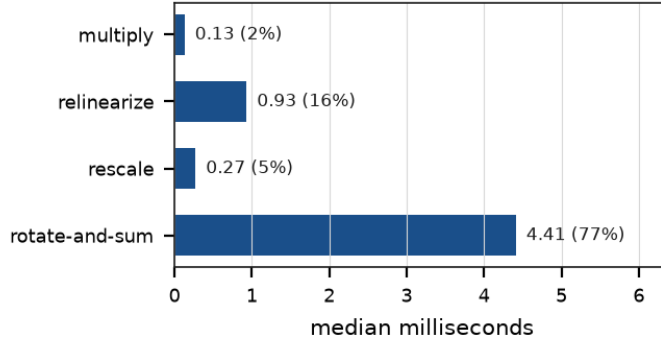


Fig. 6. Where the server spends its time inside one verification.

Table 5. Cost per verification against the number of templates packed into one ciphertext.

Batch	Server time	Per pair	Uplink each	Largest error
1	5.85 ms	5.85 ms	257.1 KiB	1.54×10^{-6}
2	5.85 ms	2.93 ms	128.6 KiB	1.56×10^{-6}
4	5.88 ms	1.47 ms	64.3 KiB	1.66×10^{-6}
8	5.86 ms	0.73 ms	32.1 KiB	1.70×10^{-6}

packing carries no cost in precision that this measurement is able to separate from run-to-run variation. At the full batch the ratio E/M is 8.2×10^{-4} , so Proposition 1 continues to hold with a factor of 1,224 in hand.

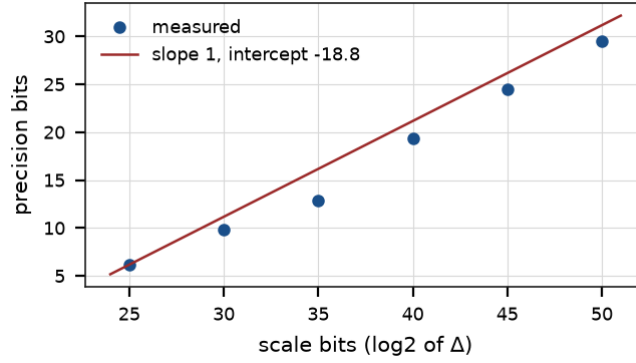
4.6 The Scaling Factor and the Minimum Precision

The sweep of Table 6 moves the scaling factor over six values while holding the circuit fixed. At a scale of s bits the chain is 60, s , 60, so the rescale divides by a prime of the same width as the scale and the identity of Section 3.2 holds at every point; the surviving prime stays 60 bits wide, and the total width of the chain runs from 145 bits at the narrowest scale to 170 at the widest, both inside the 218 that the security level permits at this degree. Server time is flat across the sweep, between 5.75 ms and 5.86 ms, because none of the operations changes.

Fitting precision against scale bits over the six points gives a slope of 0.957, a deviation of -0.043 from the value that Proposition 3 predicts, with a coefficient of determination of 0.9880 and a largest residual of 1.83 bits. The intercept is -18.83 bits, which by (3) identifies the noise floor of this circuit on this host as $\log_2 \nu = 18.83$ bits. Fig. 7 draws the measured points against the predicted slope. The agreement is worth weighing carefully, because the prediction contains nothing that could have been adjusted to produce it: the slope is fixed at one by

Table 6. Measured precision against the scaling factor. Headroom is the width of the surviving prime less the scale.

Scale bits	Headroom	Largest error	Precision	Server time
25	35 bits	1.40×10^{-2}	6.2 bits	5.86 ms
30	30 bits	1.09×10^{-3}	9.8 bits	5.86 ms
35	25 bits	1.38×10^{-4}	12.8 bits	5.76 ms
40	20 bits	1.45×10^{-6}	19.4 bits	5.75 ms
45	15 bits	4.14×10^{-8}	24.5 bits	5.75 ms
50	10 bits	1.29×10^{-9}	29.5 bits	5.75 ms

**Fig. 7.** Measured precision against scale bits, with the slope-one prediction of Proposition 3 drawn through the measured intercept.

the structure of the scheme, and the intercept is read off the data rather than chosen.

A Lower Bound on the Scaling Factor. The two results combine into a design rule. Proposition 1 requires $\varepsilon < M$, and (3) gives $\varepsilon = \nu/\Delta$, so the condition on the scaling factor is

$$\log_2 \Delta > \log_2 \nu + \log_2(1/M). \quad (6)$$

With the measured noise floor of 18.83 bits and the measured margin $M = 2.08 \times 10^{-3}$, which contributes $\log_2(1/M) = 8.91$ bits, the bound is $\log_2 \Delta > 27.7$ bits. Table 6 confirms it at the resolution of the sweep: at 30 bits the largest error is 1.09×10^{-3} , which is below M and therefore still safe, whereas at 25 bits it is 1.40×10^{-2} , which exceeds M by nearly an order of magnitude and places decisions at risk. The working point of 40 bits therefore carries 12.3 bits of slack. Those bits are not wasted, since a margin of one or two bits would leave the guarantee at the mercy of a frontend whose thresholds sit closer together,

but they are available: a deployment that needs a shorter modulus chain knows exactly how many of them it may spend, and (6) tells it how the answer changes when the embedding or the operating point changes.

5 Limitations

The system evaluated above is usable as it stands, and the points below are directions in which it would become more useful rather than obstacles to using it.

- **The client learns the numeric score.** The threshold comparison happens on the client, as Section 3.1 explains, so a deployment that must release only the accept-or-reject bit needs either a deeper circuit or a second non-colluding party. A released score is moreover a disclosure channel in its own right [1,10,14,17]. Adding noise to the score inside the ciphertext, at a scale derived from the calibration map that a matcher already fits, is a promising route to bounding that leakage without paying for a homomorphic comparison.
- **The bound uses a measured maximum.** E is the largest error over the 6,000 pairs on which decisions were taken, so Proposition 1 certifies those decisions rather than every decision the system might ever take. Replacing the measured maximum with an analytic tail bound on the circuit’s noise would extend the statement to unseen inputs, and Proposition 3 is the natural starting point because it already isolates the quantity ν that such a bound would have to control.
- **The evaluation covers one-to-one verification.** Searching a gallery of size N costs N times the work reported here before any gallery-specific encoding is applied [9]. The margin argument itself carries over unchanged, since it concerns a single comparison, but the cost figures do not, and the encodings that make such a search practical are a separate problem.
- **The impostor count bounds the rates the sample can express.** Three thousand impostor pairs resolve no false accept rate finer than 3.33×10^{-4} , and Table 3 reports which requested rates fall outside that bound. A larger impostor set is the only remedy, and it costs nothing but pairs.
- **Accuracy depends on the frontend as much as on the protocol.** The encrypted arithmetic reproduces whatever the recognition network produces, so a different network moves the accuracy figures. It leaves the cost figures untouched, and it leaves the margin argument intact, because that argument depends only on E and M , both of which are measured for whichever frontend is in use.

6 Conclusion

This paper has presented an RNS-CKKS pipeline for one-to-one face verification at multiplicative depth one, together with a way of stating what encryption costs

it. Because a decision depends on a score only through the sign of its distance to the threshold, the agreement between an encrypted matcher and its cleartext reference follows from a single inequality between the largest score error and the smallest decision margin, and a midpoint threshold grid makes that margin strictly positive by construction. On the 6,000 pairs of the Labeled Faces in the Wild pair protocol the two pipelines reach the same accuracy of 99.87% and the ratio of the two quantities is 6.07×10^{-4} , so the agreement holds with a factor of 1,647 to spare rather than by the noise that the execution happened to draw.

The same view makes the precision of the circuit a prediction instead of a measurement. One scale bit buys one precision bit, with no constant available to tune; the sweep measures a slope of 0.957 against the predicted one, and the intercept it recovers gives the noise floor. Combining that floor with the measured margin bounds the scaling factor from below at 27.7 bits, so the working point of 40 bits carries twelve bits that a deployment may trade for a shorter modulus chain. The rotation schedule of the fold is fixed by the same kind of reasoning, and its two predicted ratios, 56.8 in time and 256 in noise, are met by measurements of 57.2 and 332. One verification costs 5.75 ms of server time and 257.1 KiB of uplink, falling to 0.73 ms and 32.1 KiB when eight templates share a ciphertext.

Since the margin argument needs only two measured scalars, it applies to any encrypted matcher that releases a score and decides against a threshold, whatever its embedding, its scheme parameters or its operating point, and it tells such a system how much of its arithmetic budget it is actually using. Future work includes replacing the measured error maximum with an analytic tail bound, extending the argument from verification to gallery search, and bounding what the released score itself discloses.

References

1. Adler, A.: Sample images can be independently restored from face recognition templates. In: Canadian Conference on Electrical and Computer Engineering (2003). <https://doi.org/10.1109/CCECE.2003.1226104>
2. Albrecht, M., Chase, M., Chen, H., Ding, J., Goldwasser, S., Gorbunov, S., Halevi, S., Hoffstein, J., Laine, K., Lauter, K., Lokam, S., Micciancio, D., Moody, D., Morrison, T., Sahai, A., Vaikuntanathan, V.: Homomorphic encryption security standard. Technical report, HomomorphicEncryption.org (2018)
3. Boddeti, V.N.: Secure face matching using fully homomorphic encryption. In: IEEE International Conference on Biometrics Theory, Applications and Systems (2018). <https://doi.org/10.1109/BTAS.2018.8698601>
4. Cheon, J.H., Han, K., Kim, A., Kim, M., Song, Y.: A full RNS variant of approximate homomorphic encryption. In: Selected Areas in Cryptography, pp. 347–368 (2018). https://doi.org/10.1007/978-3-030-10970-7_16
5. Cheon, J.H., Kim, A., Kim, M., Song, Y.: Homomorphic encryption for arithmetic of approximate numbers. In: Advances in Cryptology – ASIACRYPT, pp. 409–437 (2017). https://doi.org/10.1007/978-3-319-70694-8_15
6. Cheon, J.H., Kim, D., Kim, D.: Efficient homomorphic comparison methods with optimal complexity. In: Advances in Cryptology – ASIACRYPT, pp. 221–256 (2020). https://doi.org/10.1007/978-3-030-64834-3_8

7. Deng, J., Guo, J., Xue, N., Zafeiriou, S.: ArcFace: additive angular margin loss for deep face recognition. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 4690–4699 (2019). <https://doi.org/10.1109/CVPR.2019.00482>
8. Drozdowski, P., Buchmann, N., Rathgeb, C., Margraf, M., Busch, C.: On the application of homomorphic encryption to face identification. In: International Conference of the Biometrics Special Interest Group, pp. 173–180 (2019)
9. Engelsma, J.J., Jain, A.K., Boddeti, V.N.: HERS: homomorphically encrypted representation search. *IEEE Transactions on Biometrics, Behavior, and Identity Science* **4**(3), 349–360 (2022). <https://doi.org/10.1109/TBIOM.2021.3139866>
10. Galbally, J., McCool, C., Fierrez, J., Marcel, S., Ortega-Garcia, J.: On the vulnerability of face verification systems to hill-climbing attacks. *Pattern Recognition* **43**(3), 1027–1038 (2010). <https://doi.org/10.1016/j.patcog.2009.08.022>
11. Guo, J., Deng, J., Lattas, A., Zafeiriou, S.: Sample and computation redistribution for efficient face detection. In: International Conference on Learning Representations (2022)
12. Huang, G.B., Ramesh, M., Berg, T., Learned-Miller, E.: Labeled faces in the wild: a database for studying face recognition in unconstrained environments. Technical Report 07-49, University of Massachusetts, Amherst (2007)
13. Lee, E., Lee, J.W., No, J.S., Kim, Y.S.: Minimax approximation of sign function by composite polynomial for homomorphic comparison. *IEEE Transactions on Dependable and Secure Computing* **19**(6), 3711–3727 (2022). <https://doi.org/10.1109/TDSC.2021.3105111>
14. Mai, G., Cao, K., Yuen, P.C., Jain, A.K.: On the reconstruction of face images from deep face templates. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **41**(5), 1188–1202 (2019). <https://doi.org/10.1109/TPAMI.2018.2827389>
15. Microsoft Research: Microsoft SEAL (release 4.3.3). <https://github.com/microsoft/SEAL>
- 16.atha, N.K., Chikkerur, S., Connell, J.H., Bolle, R.M.: Generating cancelable fingerprint templates. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **29**(4), 561–572 (2007). <https://doi.org/10.1109/TPAMI.2007.1004>
17. Rathgeb, C., Uhl, A.: A survey on biometric cryptosystems and cancelable biometrics. *EURASIP Journal on Information Security* **2011**, 3 (2011). <https://doi.org/10.1186/1687-417X-2011-3>
18. Zhu, Z., Huang, G., Deng, J., Ye, Y., Huang, J., Chen, X., Zhu, J., Yang, T., Lu, J., Du, D., Zhou, J.: WebFace260M: a benchmark unveiling the power of million-scale deep face recognition. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (2021). <https://doi.org/10.1109/CVPR46437.2021.01035>